

# Enterprise Web Development

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## Computing and Mathematics

May, 2026



# Enterprise Web Development (A13886)

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|---------------------|------------------------------|----------------------------|
| <b>Short Title:</b> | Enterprise Web Development   |                            |
| <b>Faculty</b>      | Science and Computing        |                            |
| <b>Department</b>   | Computing and Mathematics    |                            |
| <b>Cluster</b>      | Software and Web Development |                            |
| <b>Author</b>       | DOCONNOR                     |                            |
| <b>Credits</b>      | 10                           | <b>Level:</b> Postgraduate |

## Description of Module / Aims

To enhance the student's existing IT knowledge and experience with the skills-set and knowledge required to design, develop and deploy full-stack enterprise web applications using a range of modern platforms and tools. The module will also address foundation-level material in web architecture, object-oriented programming, design representation and relevant design patterns. In addition, the concepts and vocabulary surrounding the topics covered will enable the student to communicate and participate more effectively as a non-technical member (e.g. business analyst) of a software development team.

## Programmes

|           |                                                                    |           |
|-----------|--------------------------------------------------------------------|-----------|
| COMP-0484 | MSc in Computer Science (Enterprise Software Systems) (WD_KCESS_R) | 1 / 0 / E |
| COMP-0484 | MSc in Computing (Information Systems Processes) (WD_KISYP_R)      | 1 / 0 / E |

## Indicative Content

- Internet and Web: HTTP; Client-server architecture; Session management
- Design Patterns: Microservices; Publisher-Subscriber; MVC (Model View Controller); ORM (Object Relational Mapping)
- Web Application frameworks: In-depth case studies
- Web API Design patterns and principles: REST
- Data persistence
- Developer tool suite: Build automation; Test automation; Scaffolding; Version Control

## Learning Outcomes

*On successful completion of this module, a student will be able to:*

1. Develop a medium-sized Single Page Web App using best practice design principles and patterns.
2. Develop a web API using best practice design principles.
3. Setup and integrate suitable workflow automation tools for enterprise web development.
4. Choose particular solutions for non-functional requirements, including session management, client-server communication and security, for a user requirement set.
5. Propose and select open-source platforms and tools for the specification, documentation and testing of application components.

## Learning and Teaching Methods

- Combination of lectures and computer-based practicals.
- The lectures will cover the theory and supporting technologies behind enterprise web development.
- The lab-based practicals, building on the theoretical knowledge from lectures, provide exposure to the frameworks, tools and practical skills required to develop and build enterprise web apps.
- The practical content will use industry standard technologies, tools and techniques.

- Student will be encouraged to enhance their lab work and assessment submissions using self-directed research and learning into the state-of-the-art for enterprise web app development.

## Assessment Methods

|                       | <b>Weighing</b> | <b>Outcomes Assessed</b> |
|-----------------------|-----------------|--------------------------|
| Continuous Assessment | 100%            |                          |
| <i>Project</i>        | 60%             | 1,3,5                    |
| <i>Project</i>        | 40%             | 2,3,4                    |

## Assessment Criteria

**<40%:** Unable to interpret and apply key concepts of enterprise web app development.

**40%-59%:** Demonstrate competency in the tool suite and an ability to develop small-scale enterprise web solution.

**60%-69%:** Presents implemented solution to a medium-sized problem that demonstrate an excellent understanding of the main patterns and practices of enterprise web app design.

**70%-100%:** All of the above to an excellent standard, and incorporates self-directed investigation into state-of-the-art enterprise web technology.

## Learning Modes

| <b>Learning Mode</b> | <b>F/T Hours</b> | <b>P/T Hours</b> |
|----------------------|------------------|------------------|
| Lecture              | 24               | 24               |
| Practical            | 24               | 24               |
| Independent Learning | 222              | 222              |

## Supplementary Material

- Richardson, L. and M. Amundsen. *\_RESTful Web APIs\_*. O'Reilly Media: O'Reilly Media, 2010.

## Requested Resources

- COMPUTER LAB: BYOD Lab